

Qurui Zhang

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EDUCATION

Boston University

Master's Degree in Data Science

Boston, MA

Sep. 2025 – Present

Nanjing University of Posts and Telecommunications

Bachelor's Degree in Data Science and Big Data Technology

Nanjing, China

Sep. 2020 – June 2024

EXPERIENCE

Research Assistant

Massachusetts Institute of Technology, CSAIL, Manolis Kellis's Lab

April 2025 – Present

Boston, MA

- Architected agentic LLM pipelines automating single-cell workflows from demultiplexing to UMAP analysis
- Designed multi-agent reasoning systems integrating LLM tools for robust inference in computational biology
- Engineered provider-agnostic LLM orchestration and conversational bioinformatics platforms for domain scientists
- Developing a tree-structured agent framework in which node-level LLM agents recursively decide to split or solve tasks, while a global pipeline manages code, artifacts, and statistics for reusable AI-for-science workflows

Data Analysis Intern

SHEIN International Import & Export Co.

July 2023 – Sep. 2023

Guangzhou, China

- Built Tableau dashboards with interactive maps/filters, improving analysis efficiency by 30% for market strategy
- Analyzed sales and traffic data to flag anomalies and recommend actions that lifted new-product conversion
- Trained BiLSTM models on 1.8M reviews to extract commute insights for product design and marketing

Undergraduate NLP Research Assistant

Nanjing University of Posts and Telecommunications, HPC and Big Data Processing Institute

July 2021 – June 2023

Nanjing, China

- Developed a BiLSTM-CRF-based entity extractor with generative summarization for product reviews, capturing opinions in 77.69% of 20,000 real-world comments
- Proposed Y-Rank, a lightweight unsupervised keyphrase extraction algorithm using sentence information content and graph-based semantic similarity
- Evaluated Y-Rank on five benchmark datasets (KP20k, SemEval-2010, wikinews-fr-100, ttermith-eval, taln-archives), achieving average gains of 1.3% precision, 1.5% recall, and 1.4% F1 over baselines

PROJECTS

Colombian Extradition Data Agent | LLMs, LangChain, DeepAgent, React, Python

Sep. 2025 – Present

- Orchestrated a LangChain pipeline with DeepAgent structure to merge 3.3k+ extradition records
- Benchmarked Autogen, CrewAI, and LangChain on the extradition pipeline to compare agent orchestration and tool-call reliability, and analyzed message flows in DeepAgent System to improve inter-agent communication
- Led weekly meetings with the client and documented the full lifecycle from EDA and PoC experiments to final delivery, maintaining an audit trail of requirements, modeling choices, and dataset revisions

Culture & Tourism AI Copilot | LLMs, LangChain, RAG, LoRA fine-tuning, Python

Jan. 2024 – June 2024

- Designed a LangChain ReAct agent with RAG, four tourism tools, and a custom ChatGLM3 parser
- Fine-tuned QWEN-7B with LoRA on tourism data, raising CTKB-EVAL semantic and knowledge scores by 50%
- Evaluated the agent stack, with QWEN-14B reaching 89.3% F1 and 94.9% location accuracy

LSTM-PER-TD3 Algorithm | Deep Reinforcement Learning, PyTorch, Python

Sep. 2023 – Jan. 2024

- Proposed LSTM-PER-TD3, fusing LSTM actor-critic networks, prioritized replay, and TD3 for continuous control
- Designed memory-based actor-critics that leverage temporal history to stabilize continuous-control policies
- Implemented and evaluated LPT3 on Hopper, HalfCheetah, and Ant, where five-step histories outperformed DRL baselines in average return

TECHNICAL SKILLS

Languages: Python, SQL

Frameworks: LangChain, DeepAgent, PyTorch, React, RAG, LoRA fine-tuning

Developer Tools: Git, Docker, Tableau, Jupyter Notebook, Linux